

Omar Shawada

Operations & Process Improvement Engineer

Location: Dakahlia, Egypt — Willing to Relocate

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PROFESSIONAL SUMMARY

Operations & Process Engineer with a strong foundation in Lean Manufacturing, industrial data analysis, and digital transformation of factory operations. Proven track record of identifying operational inefficiencies, implementing structured improvement solutions, and building custom digital tools to replace manual processes. Experienced in 5S, OEE, RCFA, TPM, and Kaizen in real manufacturing environments. Background spans production planning, sheet metal operations management, and cross-functional process improvement.

KEY ACHIEVEMENTS & PROCESS IMPROVEMENTS

- **Workshop Digital Transformation — Lean + AppSheet** : Diagnosed root cause of unstructured data in a sheet metal workshop. Led 3-stage improvement: Excel (failed) → Google Sheets (partial)
* AppSheet mobile app (breakthrough). Designed Poka-Yoke interface deployed across all machines — enabling waste analysis and data-driven decisions for the first time.
- **Assembly Analyzer — Multi-Level BOM Root Cause Engine**: Built Excel-based root cause analysis tool mapping every production stoppage to exact BOM level, component, and failure type. Color-coded classification replaced guesswork with structured RCFA — enabling faster troubleshooting and clearer procurement priorities.
- **AI-Powered Production Scheduling System**: Developed dynamic scheduling app eliminating manual rescheduling bottlenecks. Supports live re-prioritization when constraints change — reducing planning cycle time significantly.
- **Supply Chain Power BI Dashboard**: Built operational visibility dashboard covering procurement gaps, production status, component issues, and delayed orders — single real-time source of truth for management.

PROFESSIONAL EXPERIENCE

Production Planning Engineer | AKL Auto Feeding Industries | Apr 2025 – Present

Process Analysis & Bottleneck Elimination

- Monitor daily production progress; identify bottlenecks and root causes using BOM-level data analysis and structured RCFA methodology.
- Maintain Assembly Analyzer tool to classify production issues by type, BOM level, and impact — enabling targeted corrective actions and eliminating cross-department investigation delays.
- Analyze procurement shortage data to proactively flag supply chain risks before impacting production; support Purchasing with prioritized action lists.

Digital Process Improvement

- Built AI-powered planning system replacing static Excel schedules with a dynamic, constraint-aware scheduling engine.
- Developed data visualization dashboard replacing fragmented manual reports with real-time operational visibility across procurement, production, and sales.
- Apply SQL queries (beginner level) to extract data from SAP ByDesign to analyze patterns, measure performance, and support continuous improvement.

Cross-functional Coordination

- Coordinate between Sales, Procurement, Engineering, and Production to resolve material constraints and maintain on-time delivery.
- Prepare structured production status and shortage reports used by management for operational decisions and resource allocation.

Sheet Metal Workshop Manager & Lean Practitioner | Independent Workshop | May 2024 – Apr 2025

Lean Manufacturing Implementation

- Applied 5S methodology to reorganize tools and workspaces — reducing tool changeover time and improving workflow efficiency.
- Implemented part nesting optimization to minimize material waste on every laser cutting job — directly reducing raw material cost per order.
- Used micro-joint techniques to optimize bending sequences and reduce bending cost and time per part.
- Standardized operations with visual management: designated storage zones, organized tool placement, clear workflow routing.

Digital Operations Transformation

- Led 3-stage digitization journey to full mobile AppSheet solution with Poka-Yoke interface — deployed on all machines regardless of computer connectivity.
- Designed Laser Cutting App (operation time, material, client tracking) and Bending App (bends, material type, thickness, quantity) — established data foundation for future OEE measurement.
- Managed workshop scheduling, job prioritization, material planning, and client delivery coordination end-to-end.

Global Industrial Management Training | 45-Hour Program | Dec 2023 – May 2024

- Maintenance Strategy Planning: Corrective, Preventive, Predictive, Proactive (ISO 14224)
- Equipment criticality analysis: FMEA, RIME methodology
- TQM data analysis tools: Pareto, Scatter, Histogram, Control Charts (Minitab & Excel)
- Root Cause Failure Analysis (RCFA): Fishbone, 5-Why
- Lean waste reduction: 7 Wastes, TPM, 5S, Kaizen, Jishu Hozen
- OEE measurement and 6 Big Losses reduction using SMED
- KPI dashboards: MTBF, MTTR, OEE, ROCE — built with Excel & data visualization tools

TECHNICAL SKILLS

Lean & CI: 5S, Kaizen, OEE, SMED, TPM, 7 Wastes, Poka-Yoke, RCFA, Fishbone, 5-Why, Visual Management

Data & Reporting: Excel, Power Query, Data Visualization, SQL (Beginner), SAP ByDesign, Minitab

AI & Digital Tools: Custom GPT, Claude AI, AppSheet, Google Sheets API, Vibe Coding

Design & Manufacturing: SolidWorks, AutoCAD, Cypcut, DXF/PDF Generation, BOM Analysis

EDUCATION

B.Sc. Mechanical Power Engineering — Mansoura University

Accumulative Grade: Good (67.33%) | 2017 – 2022

Graduation Project: Designing & Modeling Mechanical Systems for Hospital Using BIM Technology — Grade: Excellent (91.50%)

PROFESSIONAL DEVELOPMENT

- Global Industrial Management — Lean, TPM, 5S, OEE, KPIs, Maintenance Strategy (45 Hours)
- Mechanical Maintenance & Design Essentials (80 Hours)
- Process Improvement Foundations — LinkedIn Learning
- Data Analysis & Visualization — Excel, SQL, Power BI
- AI Copilot: AI Productivity & Business Automation
- 100 Essential Management Concepts — Almentor
- CPIM v8 – Preparation Program (APICS) — In Progress

LANGUAGES

Arabic: Native | English: Professional working proficiency (reading, writing, speaking)